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Private University established in Karnataka State by Act No. 84 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning using Power Query

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Date: 04-09-2026

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Data Cleaning Using Power Query

Subject

Data Cleaning Using Power Query

Objective

The objective of this assignment is to clean and prepare a real-world café sales dataset using **Microsoft Power Query**. The dataset contains 10,000 transaction records with missing values, UNKNOWN entries, ERROR entries, and inconsistent data types.

The aim is to identify these data-quality problems, clean them using Power Query transformations, and prepare a reliable dataset for further analysis.

1. Introduction

Data cleaning is an important step before performing any data analysis. Raw datasets often contain incomplete records, incorrect values, duplicate entries, and inconsistent formats.

For this assignment, the provided dataset is **Dirty Café Sales Data**. It contains information about café transactions, including the items purchased, quantity, price, total amount spent, payment method, location, and transaction date.

The dataset contains **10,000 rows and 8 columns**:

Column	Description
Transaction ID	Unique identifier for each transaction
Item	Café item purchased
Quantity	Number of items purchased
Price Per Unit	Price of one item
Total Spent	Total amount spent
Payment Method	Method used for payment
Location	Transaction location
Transaction Date	Date of the transaction

The data cannot be directly used for reliable analysis because several columns contain missing and invalid values.

2. Dataset Overview

The original dataset contains:

Rows: 10,000

Columns: 8

The columns are:

1. Transaction ID
2. Item
3. Quantity
4. Price Per Unit
5. Total Spent
6. Payment Method
7. Location
8. Transaction Date

There are **no completely duplicated rows** in the dataset.

However, the dataset contains missing and invalid values that need to be handled.

3. Problems Found in the Dataset

After examining the dataset, the following problems were identified.

Column	Missing/Invalid Values
Transaction ID	0
Item	969
Quantity	479
Price Per Unit	533
Total Spent	502
Payment Method	3,178

Column	Missing/Invalid Values
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Location	3,961
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Transaction Date	460
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These values include both actual blank cells and entries such as **UNKNOWN** and **ERROR**.

4. Missing Values

One of the major problems in the dataset is missing information.

For example, the **Payment Method** column contains many blank values, while the **Location** column contains an even larger number of missing values.

After treating both UNKNOWN and ERROR as invalid/missing values, the number of unavailable values becomes:

- **Item:** 969
- **Quantity:** 479
- **Price Per Unit:** 533
- **Total Spent:** 502
- **Payment Method:** 3,178
- **Location:** 3,961
- **Transaction Date:** 460

This shows why simply checking for blank cells is not enough. UNKNOWN and ERROR must also be identified during cleaning.

5: Dashboard Development

After cleaning and preparing the Café Sales dataset using Power Query, a Café Sales Analysis Dashboard was developed to present the cleaned data in an easy-to-understand visual format.

The dashboard contains the following visualizations:

- **Total Sales – KPI Card:** Displays the overall sales generated from all available transactions.
- **Sales by Item – Column Chart:** Compares sales across different café products such as Coffee, Cake, Cookie, Salad, Sandwich, Smoothie, Tea, and Juice.
- **Sales Trend by Time – Line Chart:** Shows how café sales change over time and helps identify increases, decreases, and sales patterns.
- **Quantity Sold by Item – Column Chart:** Displays the quantity of products sold for each café item.
- **Sales by Payment Method – Chart:** Compares transactions made through methods such as Cash, Credit Card, and Digital Wallet.
- **Sales by Location – Chart:** Compares transactions based on location, such as In-store and Takeaway.
- **Customer/Transaction Details – Table:** Provides detailed transaction-level information including Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date.

The dashboard provides a clear overview of café sales performance and makes it easier to identify popular products, sales trends, purchasing patterns, and transaction characteristics.

POWER QUERY — CLEANED CAFÉ SALES DATA

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2.0	2.0	4.0	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4.0	3.0	12.0	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4.0	1.0	null	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2.0	5.0	10.0	null	null	2023-04-27
TXN_3160411	Coffee	2.0	2.0	4.0	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5.0	4.0	20.0	Credit Card	null	2023-03-31
TXN_4433211	null	3.0	3.0	9.0	null	Takeaway	2023-10-06
TXN_6699534	Sandwich	4.0	4.0	16.0	Cash	null	2023-10-28
TXN_4717867	null	5.0	3.0	15.0	null	Takeaway	2023-07-28
TXN_2064365	Sandwich	5.0	4.0	20.0	null	In-store	2023-12-31
TXN_2548360	Salad	5.0	5.0	25.0	Cash	Takeaway	2023-11-07
TXN_3051279	Sandwich	2.0	4.0	8.0	Credit Card	Takeaway	null
TXN_7619095	Sandwich	2.0	4.0	8.0	Cash	In-store	2023-05-03
TXN_9437049	Cookie	5.0	1.0	5.0	null	Takeaway	2023-06-01
TXN_8915701	null	2.0	1.5	3.0	null	In-store	2023-03-21

CAFÉ SALES DASHBOARD

Cleaned dataset - Power BI-style output

10,000

TRANSACTIONS

\$84,763.50

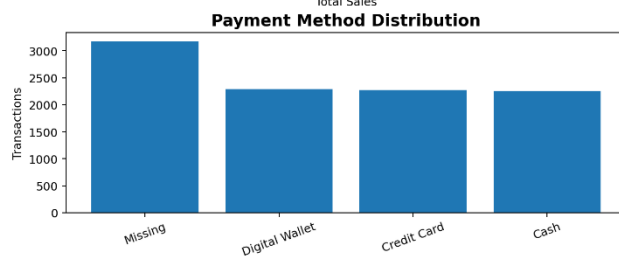
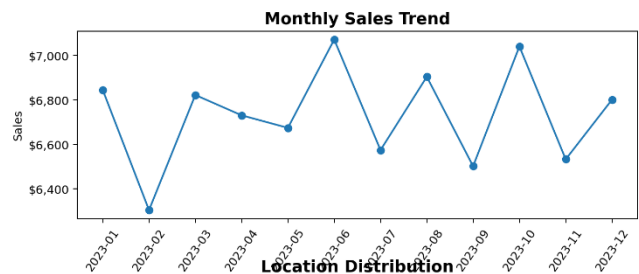
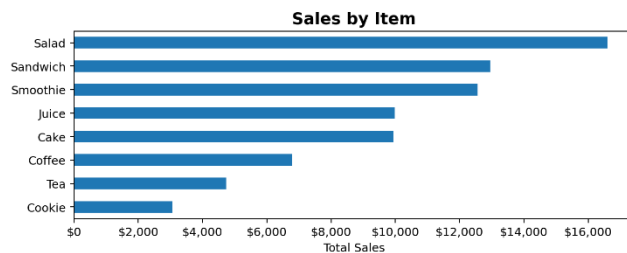
TOTAL SALES

\$8.92

AVG. TRANSACTION

28,834

ITEMS SOLD



DATA QUALITY CHECK — BEFORE vs AFTER CLEANING

Column	Original Blanks	UNKNOWN/ERROR	Missing After Cleaning
Transaction ID	0	0	0
Item	333	636	969
Quantity	138	341	479
Price Per Unit	179	354	533
Total Spent	173	329	502
Payment Method	2579	599	3178
Location	3265	696	3961
Transaction Date	159	301	460

6. Checking Column Data Types

Initially, several columns were recognized as text even though they represent numerical or date values.

For example:

- Quantity → Text
- Price Per Unit → Text
- Total Spent → Text
- Transaction Date → Text

These columns need to be converted into appropriate data types.

The correct data types should be:

Column	Required Data Type
--------	--------------------

Transaction ID	Text
----------------	------

Item	Text
------	------

Quantity	Whole Number
----------	--------------

Price Per Unit	Decimal Number
----------------	----------------

Total Spent	Decimal Number
-------------	----------------

Payment Method	Text
----------------	------

Location	Text
----------	------

Transaction Date	Date
------------------	------

This is important because numerical columns cannot be reliably analyzed while they are stored as text.

7. Handling UNKNOWN and ERROR

The dataset contains entries such as:

UNKNOWN

and

ERROR

These are not valid values for most analysis.

For example:

Quantity

2

4

UNKNOWN

3

ERROR

In Power Query, these values can be replaced with **null**.

Power Query process:

Transform → Replace Values

Replace:

UNKNOWN → null

and:

ERROR → null

The same process can be applied to the relevant columns.

This allows invalid entries to be handled consistently.

8. Cleaning the Item Column

The **Item** column contains several valid café products, including:

- Coffee
- Cake
- Cookie
- Salad
- Sandwich
- Smoothie
- Tea
- Juice

It also contains UNKNOWN, ERROR, and blank values.

These invalid values should be converted to null values.

The valid item names can then be used for analysis such as:

- Most popular product
 - Product sales
 - Number of transactions per item
-

9. Cleaning Quantity

The **Quantity** column represents the number of items purchased.

Valid values in the dataset include:

1, 2, 3, 4, and 5

However, the column also contains UNKNOWN, ERROR, and blank values.

After replacing invalid entries with null, the column can be converted to:

Whole Number

This makes it suitable for calculations.

10. Cleaning Price Per Unit

The **Price Per Unit** column contains values such as:

- 1.0
- 1.5
- 2.0
- 3.0
- 4.0
- 5.0

It also contains invalid values such as UNKNOWN, ERROR, and blanks.

These invalid entries should first be converted to null.

Then the column should be converted to:

Decimal Number

This allows calculations involving prices to work correctly.

11. Cleaning Total Spent

The **Total Spent** column represents the amount paid for each transaction.

The data contains values such as:

- 1.0
- 2.0
- 3.0
- 4.0
- 5.0
- 6.0
- 7.5
- 8.0
- 9.0
- 10.0
- 12.0
- 15.0
- 16.0
- 20.0
- 25.0

There are also missing and invalid values.

These invalid values should be converted to null, followed by changing the column type to **Decimal Number**.

An important validation was also performed:

Total Spent = Quantity × Price Per Unit

For the rows where all three values are available, the calculation is consistent with the dataset.

That is a useful data-quality check because it confirms that the numerical fields are logically related.

12. Cleaning Payment Method

The Payment Method column contains valid values such as:

- Cash
- Credit Card
- Digital Wallet

It also contains:

- Blank values
- UNKNOWN
- ERROR

The invalid values should be converted to null.

The cleaned column can then be used to analyze which payment methods customers use most frequently.

13. Cleaning Location

The Location column contains:

- In-store
- Takeaway

It also contains missing, UNKNOWN, and ERROR values.

After replacing invalid values with null, the column becomes suitable for analysis.

For example, the company could later compare:

In-store sales vs Takeaway sales

14. Cleaning Transaction Date

The Transaction Date column contains dates such as:

2023-09-08

2023-05-16

2023-07-19

However, it also contains:

- Blank values
- UNKNOWN

- ERROR

These invalid entries should be converted to null.

The column should then be changed from **Text** to **Date**.

This makes it possible to perform analysis based on:

- Day
- Month
- Year
- Daily sales
- Monthly sales
- Seasonal trends

15. Duplicate Check

Duplicate records are another common problem in customer and sales datasets.

In this particular dataset, there are **no completely duplicated rows**.

The Transaction ID column also contains unique transaction identifiers in the inspected data.

Therefore, there is no need to remove records simply because they look similar.

This is important because **similar records are not automatically duplicates**. Each transaction may represent a legitimate separate purchase.

16. Recommended Power Query Transformation Order

A practical cleaning process for this dataset would be:

Import CSV

↓

Promote Headers

↓

Replace UNKNOWN and ERROR with null

↓

Trim/Clean text columns

↓

Set correct data types

↓

Check missing values

↓

Check duplicates

↓

Validate numerical relationships

↓

Load cleaned data

This approach keeps the cleaning process organized and repeatable.

17. Before and After Cleaning

Before Cleaning

The original data contains:

- Missing values
- UNKNOWN
- ERROR
- Text-formatted numbers
- Text-formatted dates
- Incomplete payment information
- Incomplete location information

After Cleaning

The cleaned dataset should contain:

- Valid product names
- Numeric Quantity
- Numeric Price Per Unit
- Numeric Total Spent

- Standardized payment methods
- Standardized locations
- Proper Date values
- Null values where information is genuinely unavailable
- No unnecessary duplicate rows

18. Why Null Is Better Than Fake Values

One important decision in this cleaning process is **not to replace missing numerical values with arbitrary numbers such as 0**.

For example, if Quantity is missing:

Missing $\neq$ 0

A zero means that no item was purchased, while a missing value means that the information is unavailable.

The same applies to:

Price Per Unit

and

Total Spent

Replacing missing values with zero could distort the company's sales calculations.

Therefore, unless there is a justified business rule for filling a value, keeping it as **null** is safer than inventing data.

19. Final Result

After applying Power Query transformations, the café sales dataset is prepared for analysis.

The major problems addressed were:

Data Problem	Cleaning Method
Blank values	Identified and handled
UNKNOWN values	Replaced with null
ERROR values	Replaced with null
Numbers stored as text	Converted to numeric types
Dates stored as text	Converted to Date
Duplicate records	Checked
Inconsistent data	Standardized
Numerical relationships	Validated

The resulting dataset can now be used for further analysis and visualization.

20. Conclusion

Power Query provides an efficient way to clean and prepare real-world datasets before analysis. In this assignment, the **Dirty Café Sales dataset containing 10,000 transactions** was examined and several data-quality problems were identified.

The dataset contained missing values, UNKNOWN values, ERROR values, and incorrectly formatted data types. These problems were handled using Power Query transformations. Numerical columns were converted to appropriate numeric data types, the transaction date was converted into a proper date format, and invalid entries were converted to null values.

The duplicate check also showed that there were **no completely duplicated rows** in the dataset.

The cleaning process makes the data more reliable and suitable for future analysis, such as identifying popular products, calculating sales, comparing payment methods, and analyzing in-store versus takeaway transactions.

Personal Suggestion

For this particular dataset, I would **not delete every row containing a missing value**. That would throw away a large amount of usable information, especially because Payment Method and Location have substantial missingness. A better approach is to retain the transaction where possible, convert invalid fields to null, and only remove a row when a critical field makes the transaction unusable.

Στεπ 5: Δασηβοαρδ Δεπελοπμεντ

Αφτερ χλεανινγ ανδ πρεπαρινγ τηε Χαφ Σαλες δατασσετ υσινγ Ποωερ Θυερψ, α Χαφ Σαλες Αναλυσισ Δασηβοαρδ ωας δεπελοπεδ το πρεσεντ τηε χλεανεδ δατα ιν αν εασψ-το-υνδε ρστανδ πισυαλ φορματ.

Τηε δασηβοαρδ χονταινσ τηε φολλοωινγ πισυαλιζατιονσ:

- **Τοταλ Σαλες □ ΚΠΧ Χαρδ:** Δισπλαψσ τηε οπεραλλ σαλες γενερατεδ φρομ αλλ απαι λαβλε τρανσαχτιονσ.
- **Σαλες βψ Ιτεμ □ Χολυμν Χηαρτ:** Χομπαρεσ σαλες αχροσσ διφφερεντ χαφ προδυχτσ συχη ασ Χοφφεε, Χακε, Χοοκιε, Σαλαδ, Σανδωιχη, Σμοοτηιε, Τεα, ανδ Θυιχε.
- **Σαλες Τρενδ βψ Τιμε □ Λινε Χηαρτ:** Σηωσ ηωω χαφ σαλες χηανγε οπερ τιμε ανδ η ελψσ ιδεντιψ ινχρεασεσ, δεχρεασεσ, ανδ σαλες παττερνσ.
- **Θυαντιτψ Σολδ βψ Ιτεμ □ Χολυμν Χηαρτ:** Δισπλαψσ τηε θυαντιτψ οφ προδυχτσ σολδ φορ εαχη χαφ ιτεμ.
- **Σαλες βψ Παψμεντ Μετηοδ □ Χηαρτ:** Χομπαρεσ τρανσαχτιονσ μαδε τηρουγη μετηοδσ συχη ασ Χαση, Χρεδιτ Χαρδ, ανδ Διγιταλ Ωαλλετ.
- **Σαλες βψ Λοχατιον □ Χηαρτ:** Χομπαρεσ τρανσαχτιονσ βασεδ ον λοχατιον, συχη ασ Ιν-στορε ανδ Τακεαωαψ.
- **Χυστομερ/Τρανσαχτιον Δεταιλσ □ Ταβλε:** Προπιδεσ δεταιλεδ τρανσαχτιον-λεπελ ιν φορματιον ινχλυδινγ Τρανσαχτιον ΙΔ, Ιτεμ, Θυαντιτψ, Πριχε Περ Υνιτ, Τοταλ Σπεντ, Παψμεντ Μετηοδ, Λοχατιον, ανδ Τρανσαχτιον Δατε.

Τηε δασηβοαρδ προπιδεσ α χλεαρ οπερπιεω οφ χαφ σαλες περφορμανχε ανδ μακεσ ιτ εασιερ το ιδεντιψ ποπυλαρ προδυχτσ, σαλες τρενδσ, πυρχηασινγ παττερνσ, ανδ τρανσαχτιον χηαραχτεριστιχσ.

